

Axial Turbine Theory and Design

Class 6

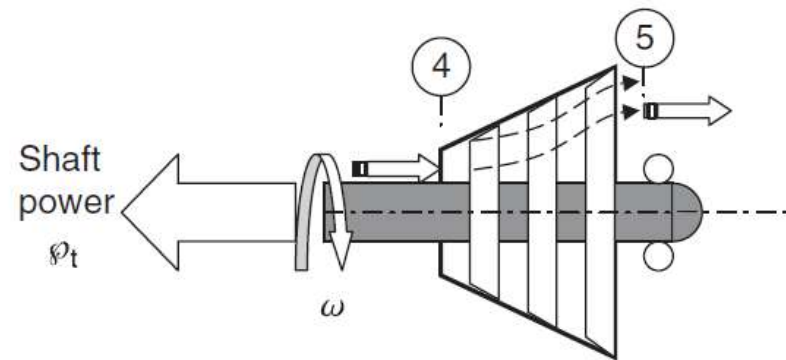
Introduction

- In this class, we will introduce aerodynamics of turbine blades, optimal nozzle exit tangential Mach number, optimum solidity, turbine losses, and cooling.

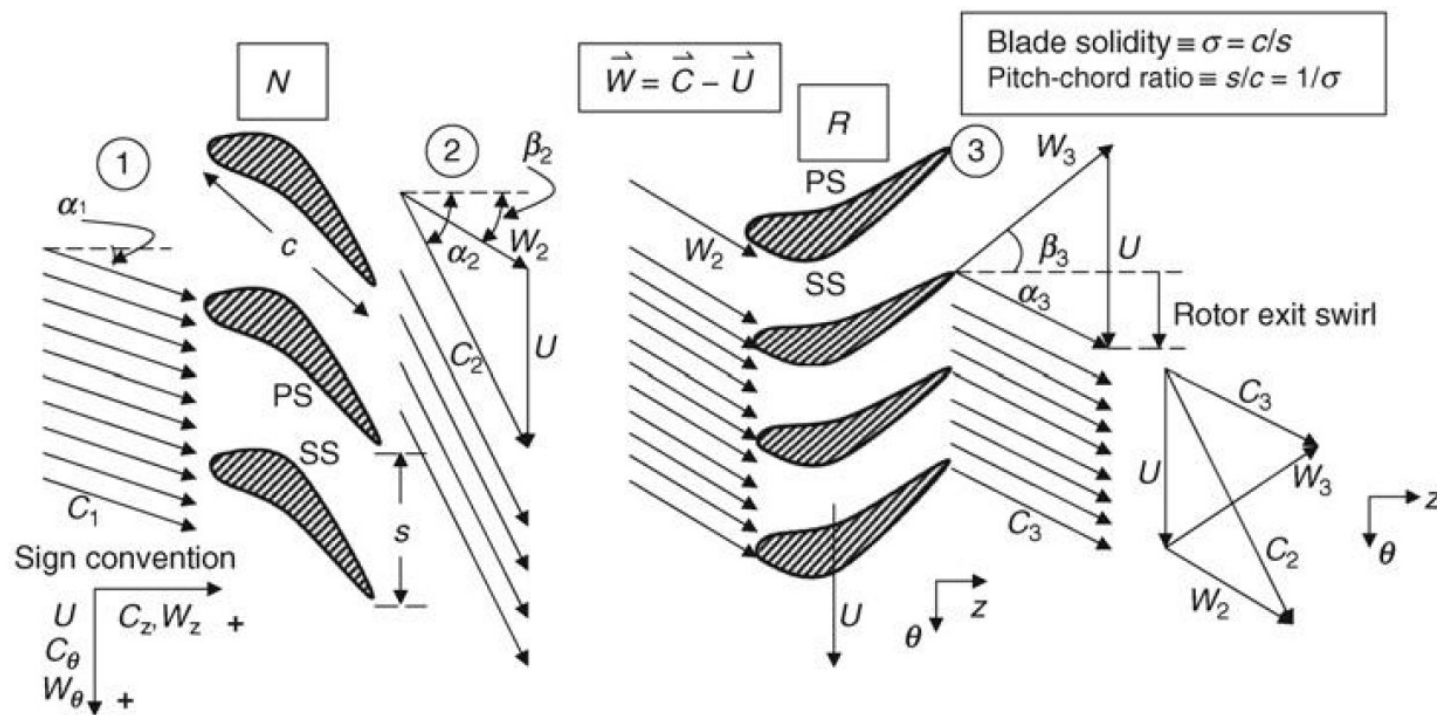
The Geometry

The notion that a turbine is a “mirror image” of a compressor is misleading. The evidence of many compressor stages (~5 in modern gas turbine engines of the same mass flow rate) being driven by a single turbine stage is a compelling reason for the inherent differences in the fluid mechanics of the two devices. The challenges in a turbine are primarily in the cooling techniques and their effectiveness;

tip clearance control, blade life, and material characteristics suitable for a high-temperature environment and corrosion. In the compressor, the flow stability (rotating stall and surge) and efficiency are highly critical to the design.



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Rothalpy

- Definition
 - Rotational enthalpy (relative enthalpy through a rotor)

$$I = h_t - W^2/2$$

- Conservation of Rothalpy
 - This states that the relative enthalpy through a rotor blade row remains constant even for a change of radius, therefore

$$I_2 = I_3$$
$$h_2 - W_2^2/2 = h_3 - W_3^2/2$$

this could also be expressed as

$$T_2 - T_3 = \frac{(W_3^2 - W_2^2)}{2}$$

Stage Reaction

The choice of the degree of reaction in a turbine ranges from zero to upward of 50%. The degree of reaction in a turbine is defined as the ratio of static enthalpy drop in the rotor to that of the stage, namely

$$^{\circ}R \equiv \frac{h_2 - h_3}{h_1 - h_3} = \frac{h_{t2} - h_{t3} - (C_2^2 - C_3^2)/2}{h_{t1} - h_{t3} - (C_1^2 - C_3^2)/2}$$

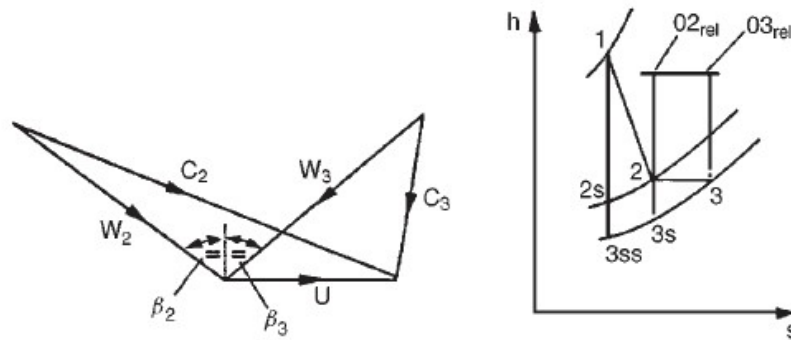
Assuming the inlet and exit kinetic energies of the stage are equal, the above expression is simplified to

$$^{\circ}R = 1 - \frac{C_2^2 - C_3^2}{2(h_{t2} - h_{t3})} = 1 - \frac{C_{\theta 2}^2 - C_{\theta 3}^2}{2U(C_{\theta 2} - C_{\theta 3})} = 1 - \frac{C_{\theta 2} + C_{\theta 3}}{2U}$$

Choice of Reaction

Zero degree of reaction

- A zero degree of reaction turbine stage, produces the entire static enthalpy drop of the stage across the nozzle. The rotor produces no static enthalpy drop.

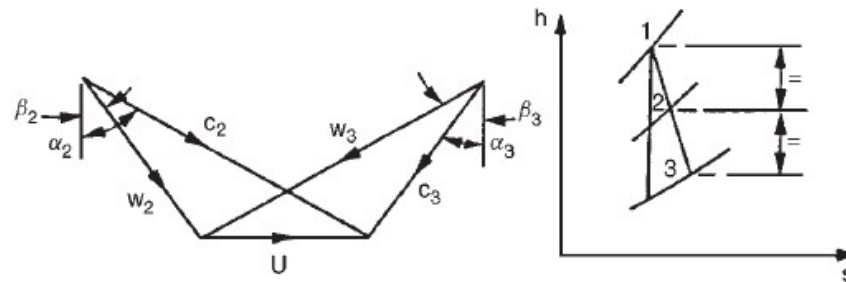


Velocity diagram and $h-s$ diagram for a 0° reaction turbine stage.

Choice of Reaction

50% reaction stage

- The combined velocity diagram for the 50% reaction stage is symmetrical meaning $\beta_3 = \alpha_2$. Because of the symmetry it is at once obvious that $\beta_2 = \alpha_3$. The 50° degree reaction implies that the enthalpy drop in the nozzle row equals the enthalpy drop in the rotor, or $h_1 - h_2$

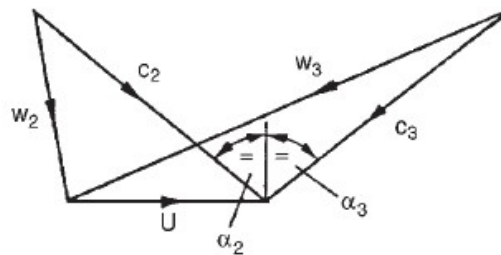


Velocity diagram and $h - s$ diagram for a 50% reaction turbine stage.

Choice of Reaction

100% reaction stage

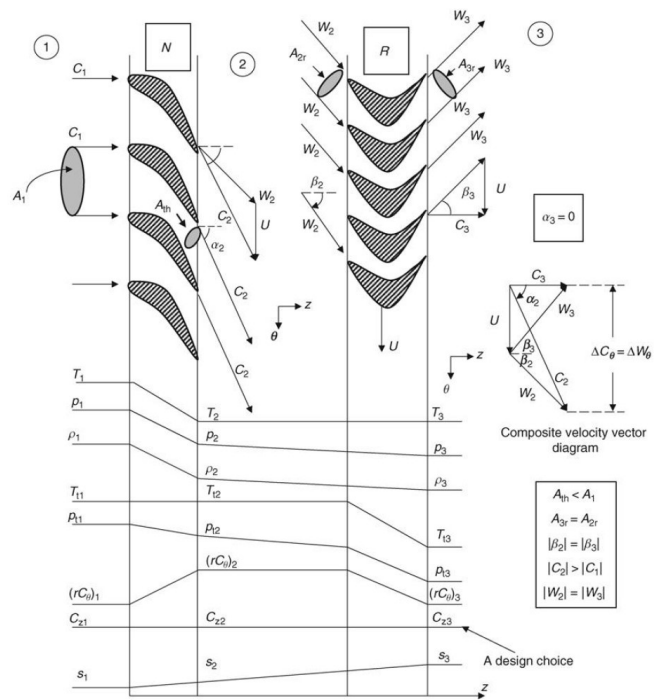
- For a 100% reaction stage (where all the work is extracted on the rotor row) therefore $\alpha_3 = \alpha_2$ and also $c_2 = c_3$. It is important to understand the nature of the Flow within a 100% reaction stage, as it becomes apparent that if R exceeds unity, then $c_2 < c_1$ (nozzle flow



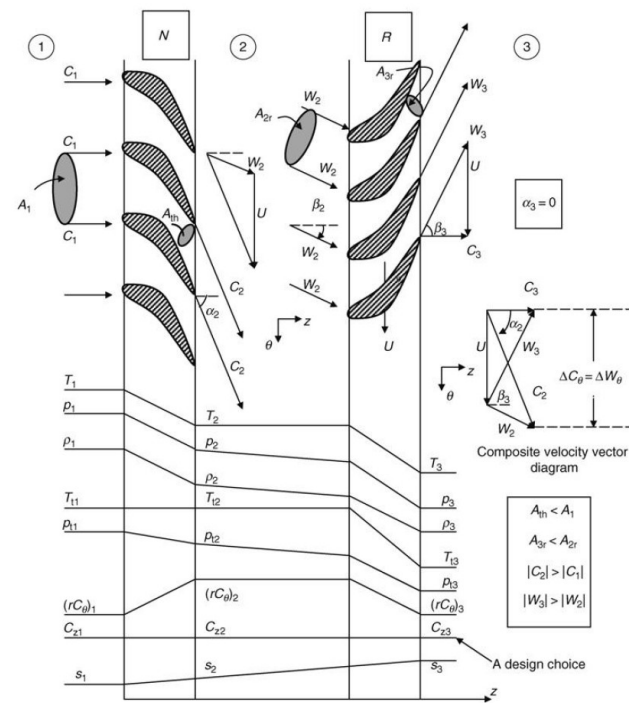
Velocity diagram and $h - s$ diagram for a 100% reaction turbine stage.

Choice of Reaction

0° Reaction Turbine



50° Reaction Turbine



Choice of Axial Velocity

- For an HP turbine annulus geometry and for other turbine stages downstream, the axial annulus area distribution will be defined by the choice of distribution of axial velocity.
 - Constant C_z
 - Falling C_z
 - Rising C_z

Which parameter will be directly impact by your design selection?

Choice of Axial Velocity - Rising Cz

- HP Turbines
 - When compared with 'constant Cz', a choice of 'rising Cz' would result in higher blade friction losses and, therefore, lower efficiency. At the same time, however, blade height would be lower leading to reduced blade mass, lower stress for a given RPM and less cooling air requirement for cooled stages.
- LP Turbines
 - When compared with 'constant Cz', a choice of 'rising Cz' would result in higher blade friction losses and, therefore, lower efficiency. Simultaneously, the turbine exit velocity increases giving rise to increased exhaust friction losses or a longer exhaust diffuser. At the same time, however, blade height would be lower, leading to reduced blade mass, lower blade stress and lower rim load (AN^2) for a given RPM.

Choice of Axial Velocity - Falling Cz

- HP Turbines

- When compared with 'constant Cz', a choice of 'falling Cz' would result in lower blade friction losses and, therefore, higher efficiency. At the same time, however, blades would be longer with increased blade mass, higher stress for a given RPM and more cooling air requirement for cooled stages.

- LP Turbines

- When compared with 'constant Cz', a choice of 'falling Cz' would result in lower blade friction losses and, therefore, higher efficiency. Simultaneously, the turbine exit velocity decreases giving rise to reduced exhaust friction losses or a shorter exhaust diffuser. At the same time, however, blade height would be higher, with increased blade mass, higher blade stress and higher rim load (AN^2) for a given RPM.

Utilization Factor

- In a turbine, not all energy supplied can be converted into useful work - even with an ideal fluid. There must be some kinetic energy at the exit that is discharged due to the exit velocity. Thus, the utilization factor is defined as the ratio of ideal work to the energy supplied

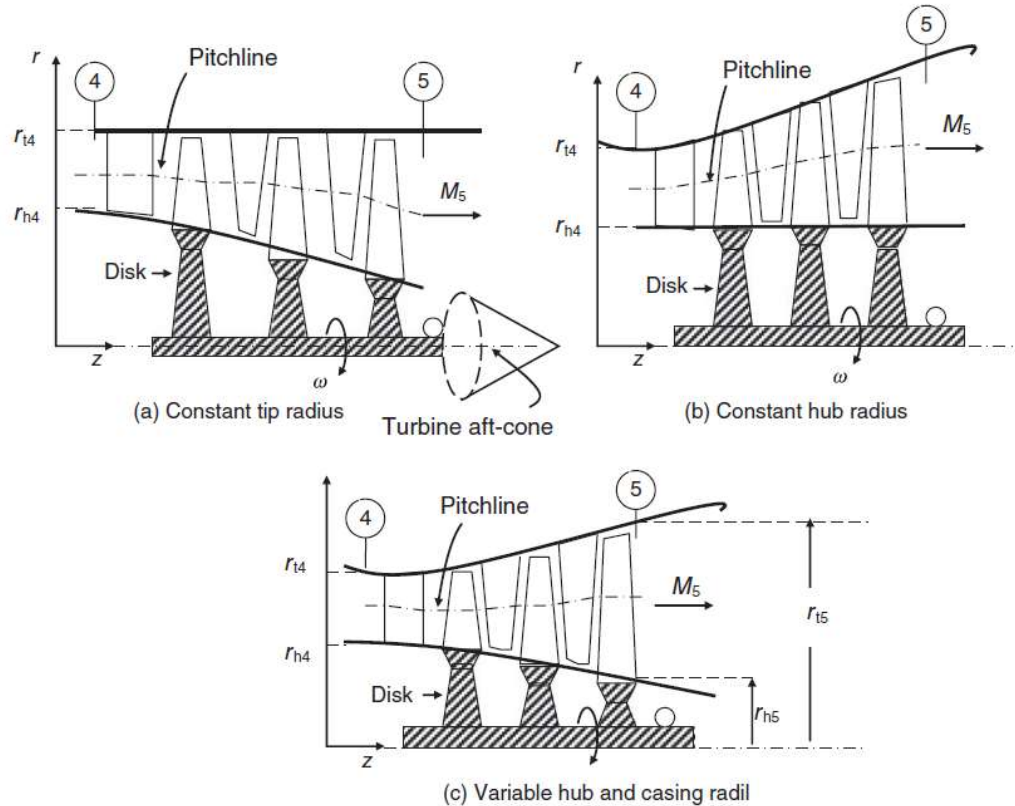
$$E = \frac{H_{id}}{H_{id} + \frac{C_{exit}^2}{2g}}$$

- and it can be written in terms of the velocity for a single rotor with constant radius

$$E = \frac{(C_{in}^2 - C_{out}^2) + (W_{out}^2 - W_{in}^2)}{C_{in}^2 + (W_{out}^2 - W_{in}^2)}$$

Annulus Design or Choice

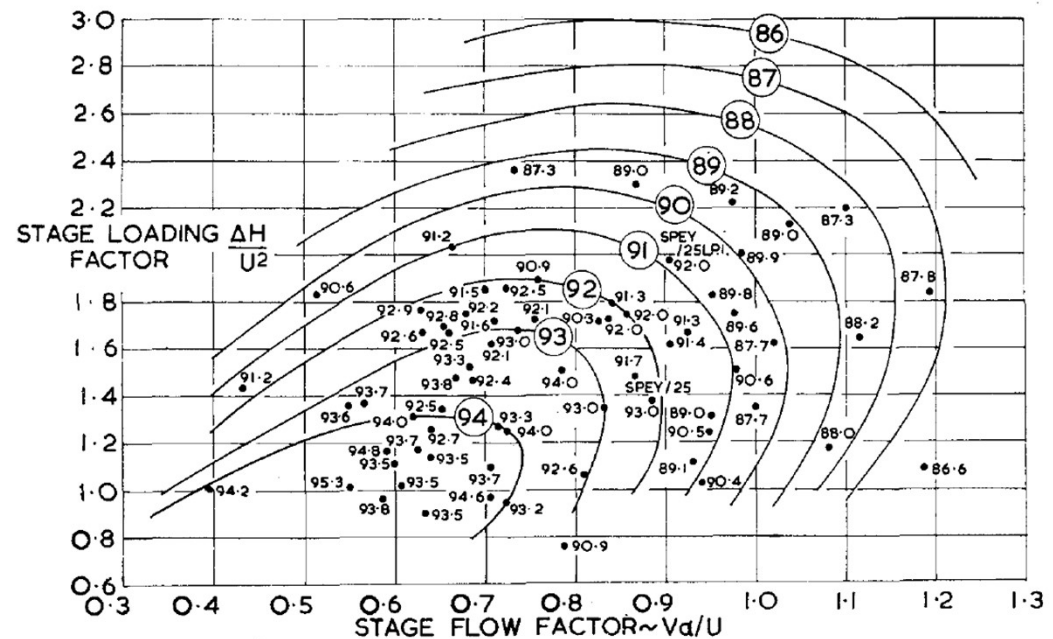
- The annulus sizing and the geometry are directly affected by our choice of the axial velocity and the mass flow rate through continuity equation. With a constant axial velocity and a decrease in gas density in the flow direction, the annulus flow area needs to grow inversely proportional to the density drop.
- In the case of a cooled turbine, the coolant mass flow rate needs to be added to the hot gas flow rate in establishing the annulus flow area, via continuity equation.



Turbine Blading Efficiency

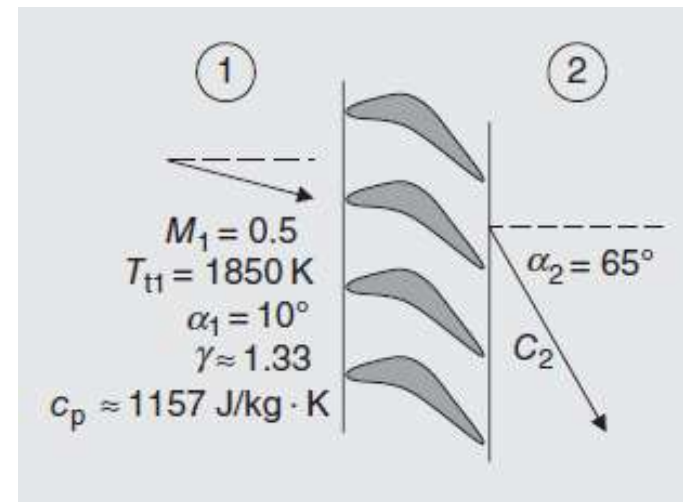
- The Smith Chart

- For turbine blading, it is assumed that the blade shapes are determined within close limits to the calculated flow angles. It is therefore possible, from the results of cascade tests on a family of blades, to predict the losses in the blade rows and have an estimate for stage efficiency. One such method to estimate stage efficiency is shown by the efficiency contours superimposed on a range of stage flow and loading coefficients, ϕ and ψ . These efficiency contours are based on gas turbine designs up to 1965 with stage reaction of 50%, and efficiency values corrected for zero tip clearance.
- The following conclusions may be deduced. For aeroengines, the optimum value for ψ is from 1.5 to 2.5 and for ϕ is from 0.8 to 1.2. The efficiency ranges from 86% to 91%. For industrial gas turbines, the optimum value for ψ ranges from 1.0 to 1.5, while ϕ ranges from 0.4 to 0.8. The corresponding efficiency ranges from 93% to 95%.



Example 10.1

- The combustor discharge into a turbine nozzle has a total temperature of 1800 K and axial Mach number of 0.55, as shown.
- Assuming that the nozzle is uncooled, the axial velocity remains constant across the nozzle, and the absolute flow angle at the nozzle exit is $\alpha_2 = 60^\circ$, calculate:
 - inlet velocity C_1 in m/s
 - the exit absolute Mach number M_2
 - nozzle torque per unit mass flow rate for $r_1 \approx r_2 = 0.35$ m

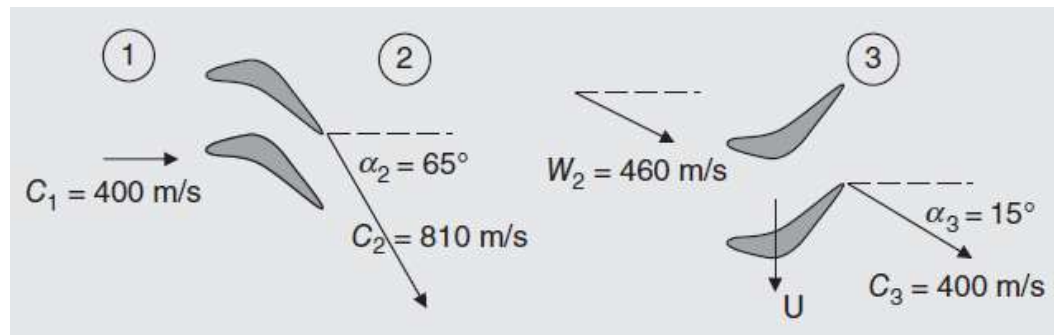


Example 10.2

- The turbine nozzle in Example 10.1 produced a slightly supersonic exit flow with $M_2 \approx 1.19$. If we turn the flow to a reduced α_2 level, we may reduce the exit Mach number M_2 . Calculate the nozzle exit flow angle α_2 if we wish the exit Mach number to be $M_2 \approx 1.0$, i.e., choked.

Example 10.3

- A turbine stage at pitchline has the following velocity vectors, as shown.



- Calculate
 - a) the axial velocities up- and downstream of the rotor
 - b) the rotor velocity U
 - c) the degree of reaction at this radius.

Nozzle Exit Swirl

- The effect of the degree of reaction on the mass flow rate and optimal swirl in a turbine stage, for a given blade tangential Mach number M_T , is considered in this section. For simplicity of the expressions, we assume that the exit flow from the rotor is swirl free, that is, $C_{\theta 3} = 0$.

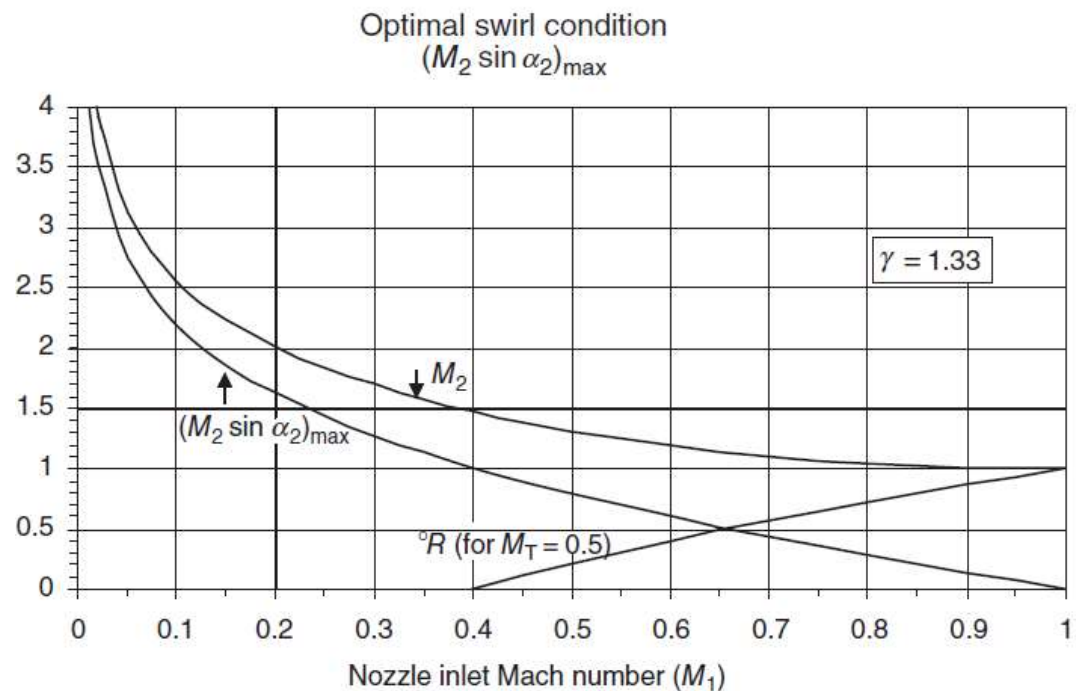
$$M_2 \sin \alpha_2 = M_{\theta 2} = 2M_T(1 - \sigma R)$$

- Assuming the total temperature, total pressure and flow area remain constant across the turbine nozzle, and satisfying the continuity equation, the equation for maximizing the nozzle exit swirl Mach number is as follows:

$$(M_2 \sin \alpha_2)^2 = M_2^2 - M_1^2 \left[\frac{1 + (\gamma - 1) M_2^2 / 2}{1 + (\gamma - 1) M_1^2 / 2} \right]^{\frac{\gamma+1}{\gamma-1}}$$

Nozzle Exit Swirl

- Under optimal conditions ($M_T=0.5$ & $\gamma=1.33$), a choice of the stage degree of reaction establishes a turbine inlet Mach number, M_1 .
- An impulse turbine stage, with $\circ R = 0$, demands a nozzle inlet Mach number of ~ 0.4 and a 50% reaction stage corresponds to an inlet Mach number ~ 0.65 .
- An impulse stage is thus capable of producing maximum specific work, but at a reduced mass flow density. The optimal exit Mach number M_2 that maximizes the swirl and stage work, then varies between 1.2 and 1.5 for a stage reaction that varies between 50% and 0, respectively.



Blade Loading

- The blade loading per unit span is the integral of the pressure difference on the pressure and suction surfaces, which produces the blade tangential force F_θ .

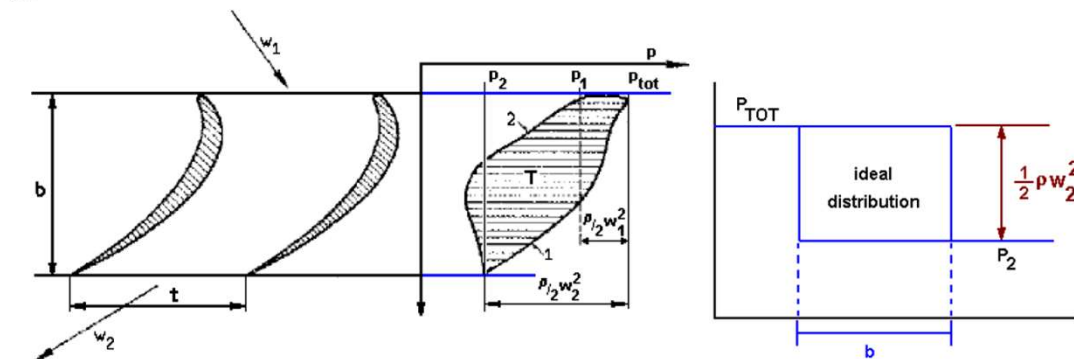
$$F_\theta = c_z \int_0^1 (p_p - p_s) d(z/c_z)$$

- Also from the momentum balance across the blade row in the tangential direction, we get the blade tangential force as

$$F_\theta = \rho_2 S \cdot W_{z2} (W_{\theta 1} - W_{\theta 2})$$

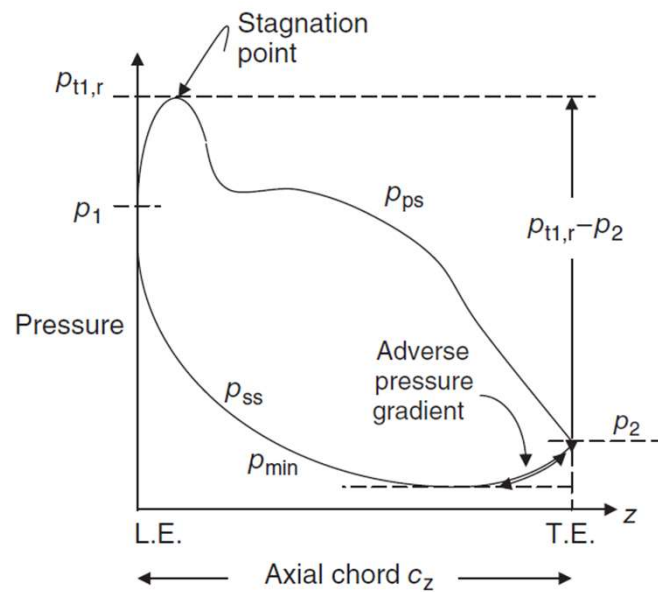
noting that

$$W_{\theta 1} - W_{\theta 2} = C_{\theta 1} - C_{\theta 2}$$

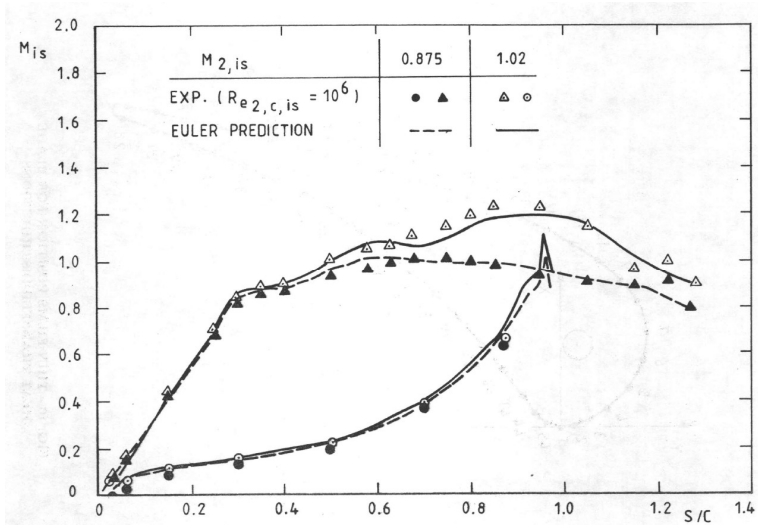


Blade Loading

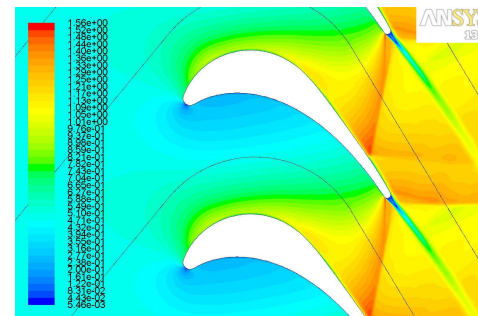
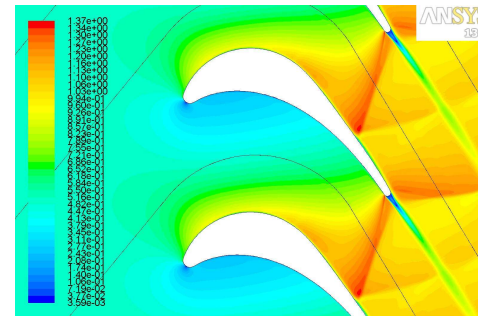
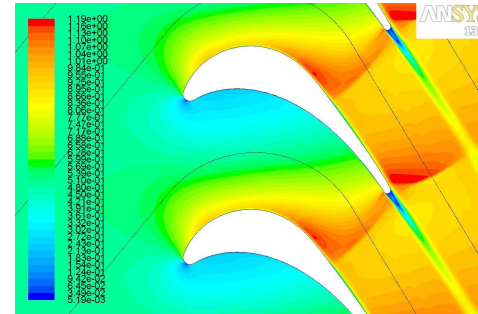
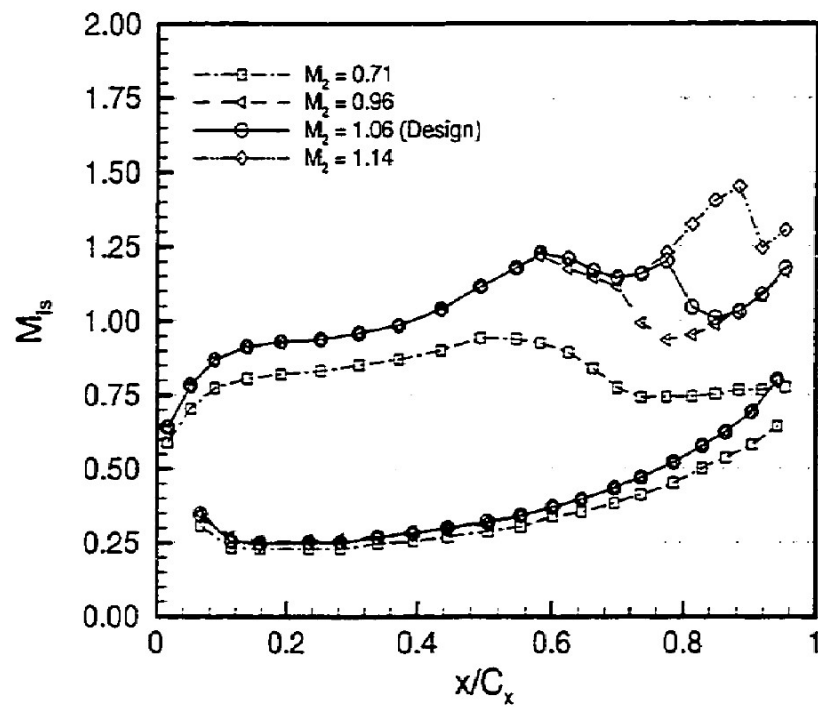
Blade Loading as a function of pressure



Blade Loading as a function of Mach No.



Blade Loading



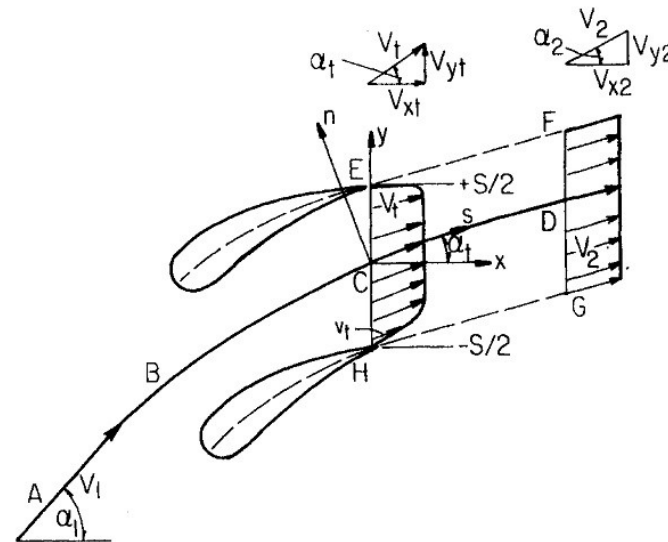
Turbine Blade Losses (Loss Mechanisms)

- We may categorize the losses in a turbine blade row as having their origins in
 1. Profile loss, which is established based on 2D cascade studies
 2. Secondary flow losses, which may be estimated using secondary flow theory of Hawthorne or the approximate version by Squire and Winter
 3. Annulus losses, which includes tip clearance loss, (seal) leakage loss, casing boundary layer loss, corner vortex loss
 4. Coolant-related losses, including mixing loss, induced separation loss
 5. Shock and shock-boundary layer interaction losses in transonic stages of LPT
 6. Unsteady flow loss causing shock oscillation and vortex shedding in the wake

Turbine Blade Losses

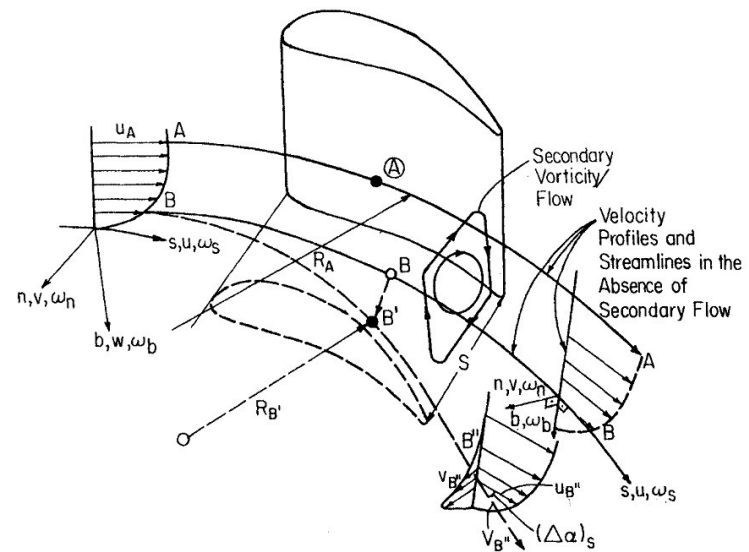
- Blade Profile Loss

- This term encompasses all the losses due to blade boundary layers and wakes through viscous as well as turbulent dissipation.
- The dissipation of mechanical energy into heat within the boundary layer results in an entropy increase inducing stagnation pressure loss even though the stagnation enthalpy is known to be constant for an adiabatic flow. The trailing vortex systems in the blade wake, their eventual mixing and later dissipation give rise to additional losses. Regarding the three dimensional effects, these losses depend on the radial distribution of the pressure gradient as well as the three dimensional blade geometry, including blade leaning, dihedral, sweep, twist, aspect ratio and meridional curvature



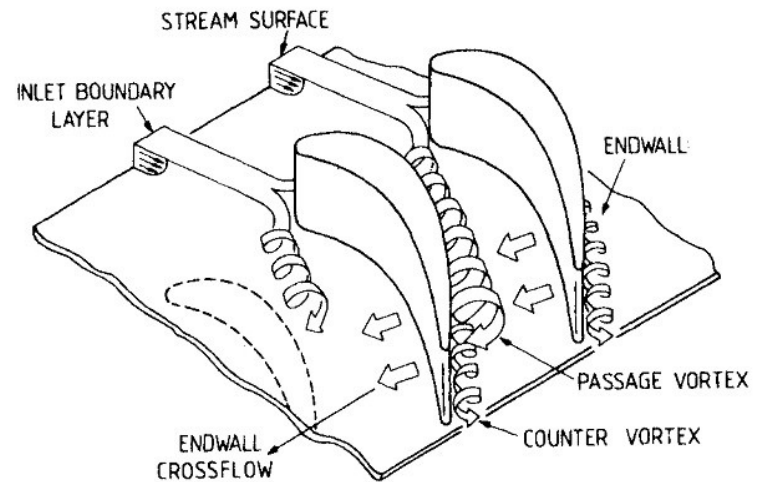
Turbine Blade Losses

- Secondary Flow Losses
 - The secondary flow and the formation of vortex produce mixing and dissipation of energy similar to the phenomena described under clearance losses. An indirect effect of the secondary flow losses is the initiation of corner stall, an interaction of secondary flow with the wall and the blade boundary layer, and the wake result in additional losses. Secondary flow losses depend on blade loading, Reynolds number, Mach number and boundary layer thickness, among other variables.



Turbine Blade Losses

- Annulus/Endwall Flow Losses
 - In this category losses are due to annulus and hub-wall boundary layers, which are almost invariable and three dimensional in nature. Depending on the type of turbomachinery, the endwall losses may represent approximately 50 to 70% of total losses. This type of loss mechanism is also due to the corner separation, induced by adverse pressure gradient and secondary flow.



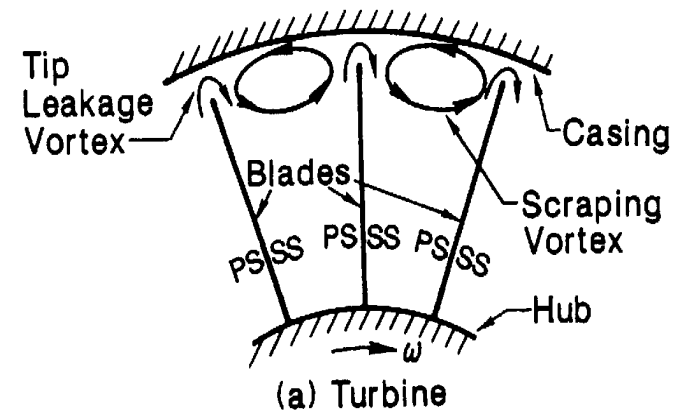
Turbine Blade Losses

- Shock Losses
 - Viscous dissipation across the shock leads to an increment in the losses. The sudden increase in static pressure across the shock results in thickening of the boundary layer and/or flow separation. Depending on the Mach and Reynolds number, the shock loss can represent a considerable part of total profile losses. The three dimensional blade geometry and radial variation of properties, including the inlet Mach number, have a major influence on radial variation of this type of losses.



Turbine Blade Losses

- Tip Leakage Losses
 - Leakage flow in the tip clearance of the rotor or axial gaps in the flow path are the main causes of tip leakage losses. The main factors influencing the tip leakage and the tip clearance are the gap size, the cascade incidence as well as the pressure and suction surface. The loss is mainly caused by the different angles and flows between the main flow and the leakage flow near the suction surface and even downstream of the blade, where the dissipation accounts for losses. In addition, the formation of a leakage vortex, its eventual dissipation and interaction with the main flow can result in energy costs.

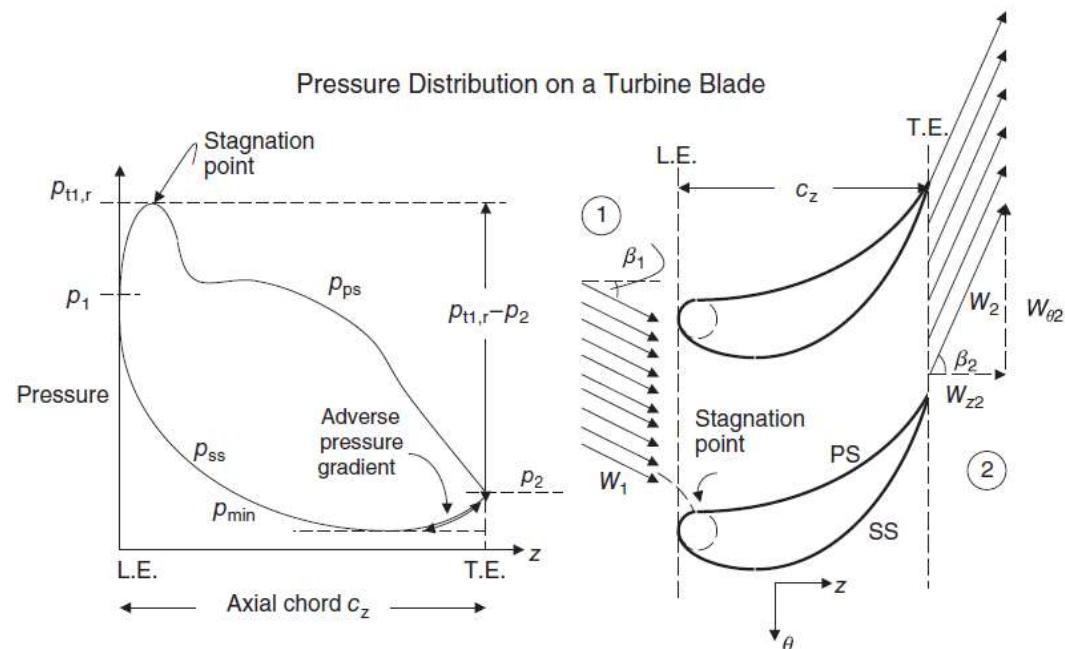


Turbine Blade Losses

- Different methods, correlations and approximations have been studied and proposed to assess the stage performance and account for the different turbine loss sources.
- Some examples but not limited to are:
 - 1) Ainley, D. and Mathieson, G. (1951), *A method of performance estimation for axial-flow turbines*, Defense Technical Information Center.
 - 2) Craig, H. and Cox, H. (1970), "Performance estimation of axial flow turbines", *Proceedings of the Institution of Mechanical Engineers*, vol. 185, no. 1, pp. 407-424.
 - 3) Denton, J. D. (1993), "The 1993 IGTI Scholar Lecture: loss mechanisms in turbomachines", *Journal of Turbomachinery*, vol 114, no. 4 pp. 221-256.
 - 4) Benner, M. W., Sjolander, S. A. and Moustapha, S. H. (2006), "An empirical prediction method for secondary losses in turbines - Part I: A new loss breakdown scheme and penetration depth correlation", *Journal of Turbomachinery*, vol. 128, no. 2, pp. 273-280.
 - 5) Benner, M. W., Sjolander, S. A. and Moustapha, S. H. (2006), "An empirical prediction method for secondary losses in turbines - Part II: A new secondary loss correlation", *Journal of Turbomachinery*, vol. 128, no. 2, pp. 281-291.

Solidity

- Although the flow in turbines is characterized as having favorable pressure gradient, the flow acceleration on the suction surface of a turbine blade has to face an adverse pressure gradient toward the trailing edge. This behavior is typical of all lifting surfaces whether a compressor or a turbine blade.



Optimum Solidity

- Zweifel (1945) has addressed the issue of optimum solidity in turbomachinery by introducing a loading parameter and a criterion that is widely accepted. Here we present Zweifel's methodology. Equating the two expressions for the blade tangential force (per unit span) and isolating axial solidity c_z/s yields

$$\sigma_z \equiv \frac{c_z}{s} = \frac{\rho_2 W_{z2} (W_{\theta 1} - W_{\theta 2})}{\int_0^1 (p_p - p_s) d\left(\frac{z}{c_z}\right)}; \quad \psi_z \equiv \frac{\int_0^1 (p_p - p_s) d\left(\frac{z}{c_z}\right)}{p_{t1,r} - p_2}$$

- Now, in terms of the Zweifel loading parameter,

$$\sigma_z \equiv \frac{\rho_2 W_{z2} (W_{\theta 1} - W_{\theta 2})}{(p_{t1,r} - p_2) \psi_z}$$

Optimum Solidity

- Zweifel recommended a value of 0.8 (to ~ 1) for the loading parameter ψ_z . Hence, based on the exit flow angle and swirl ratio across the blade row, an optimum solidity is obtained with a maximum for β_2 of 45° for a blade row with a fixed swirl ratio.

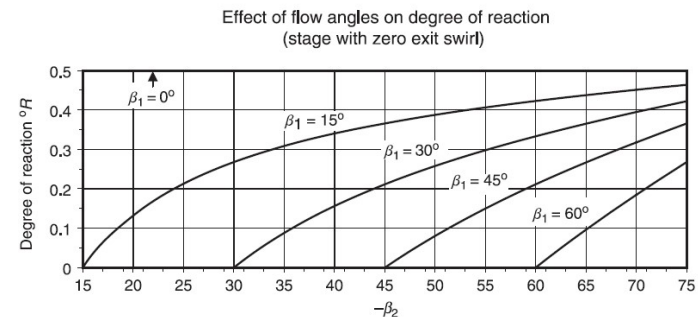
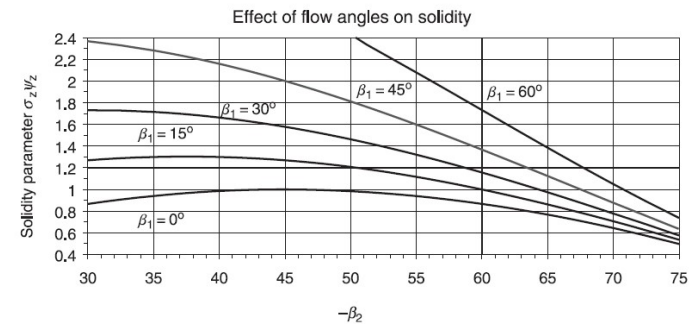
$$\sigma_z \equiv \left(\frac{(\gamma/2)M_{2r}^2}{\left(1 + \frac{\gamma-1}{2}M_{2r}^2\right)^{\frac{\gamma}{\gamma-1}} - 1} \right) \frac{\left(\frac{W_{\theta 1}}{W_{\theta 2} - 1}\right) \sin(2\beta_2)}{\psi_z}$$

- In the incompressible limit (i.e., $M_{2r} \rightarrow 0$), assuming constant axial velocity, and trigonometric relations, the above expression simplifies to

$$\sigma_z \psi_z = \frac{2 \cos \beta_2}{\cos \beta_1} \sin(\beta_1 - \beta_2)$$

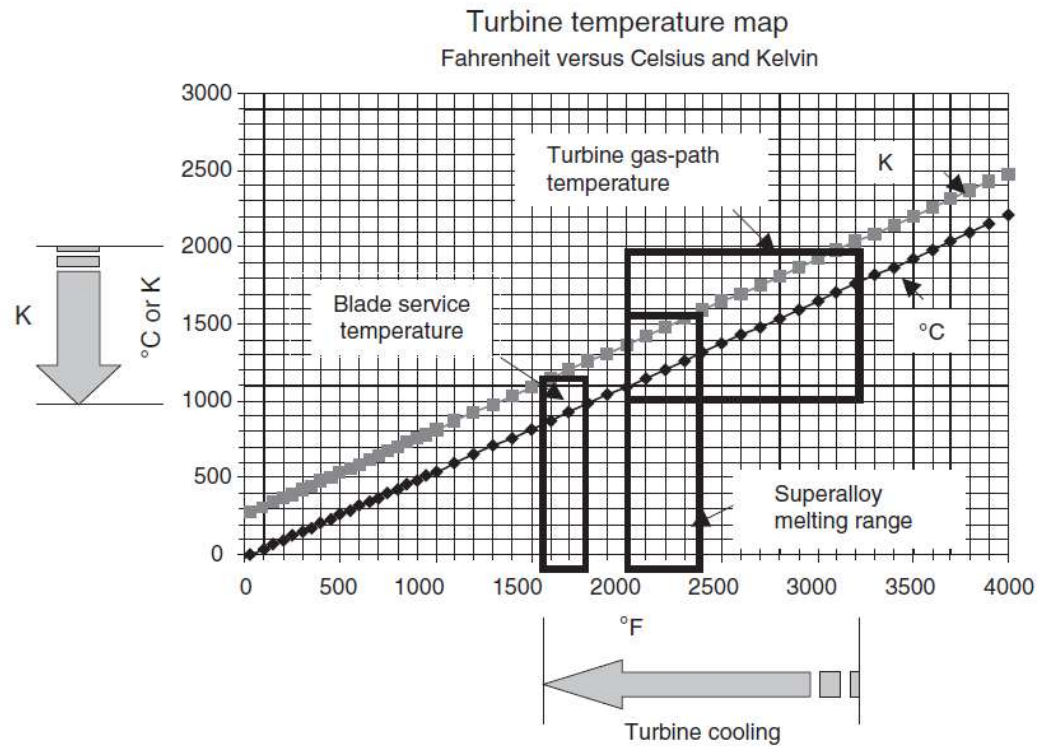
Optimum Solidity

- In the region of interest in a turbine, that is, for the exit flow angles between -45° and -75° , we observe that the solidity parameter decreases with the decrease in exit flow angle for a given inlet flow angle.
- Also, for a given exit flow angle $-\beta_2$, the increase in the inlet flow angle causes a lower stage reaction (i.e., moving toward impulse stage) and thus an increase in blade loading.



Turbine Cooling

- The entrance flow to a turbine nozzle is the combustor exit flow with its spatially nonuniform characteristics, such as the hot spots.
- The average combustor exit temperature in modern gas turbine engines at takeoff is ~1750–2000 K and the temperature nonuniformity or “hot spots” may expose the nozzle to an additional temperature of 100–200 K.
- This is clearly beyond the temperature limits of metals, including the nickel-based alloys used in turbines.



Turbine Cooling Effectiveness

- A cooling effectiveness parameter η may be defined as:

$$\eta \equiv \frac{T_{tg} - T_{wg}}{T_{tg} - T_c}$$

which may be used as a figure of merit in modern gas turbine engines where it attains a value between 0.6 and 0.7.

- The turbine deals with several temperatures in a cooled turbine, these are:
 1. T_g , which is the gas static temperature
 2. T_{aw} is the adiabatic wall temperature, that is, the temperature of the insulated wall
 3. T_t , the total gas temperature in the absolute frame, for example, in the nozzle or casing frame of reference
 4. T_{tr} , the total gas temperature as seen by the turbine rotor
 5. T_{wg} is the wall temperature on the hot gas side (i.e., actual or desired surface temperature)
 6. T_{wc} is the wall temperature on the coolant side
 7. T_c is the coolant temperature,

Turbine Cooling Effectiveness

- The adiabatic wall temperature in gas turbine is very nearly equal to the total or stagnation temperature of the gas. For a flat plate, the adiabatic wall temperature is

$$T_{aw} = T_g + r \frac{C^2}{2C_p} \text{ for stator}$$

$$T_{aw,r} = T_g + r \frac{W^2}{2C_p} \text{ for rotor}$$

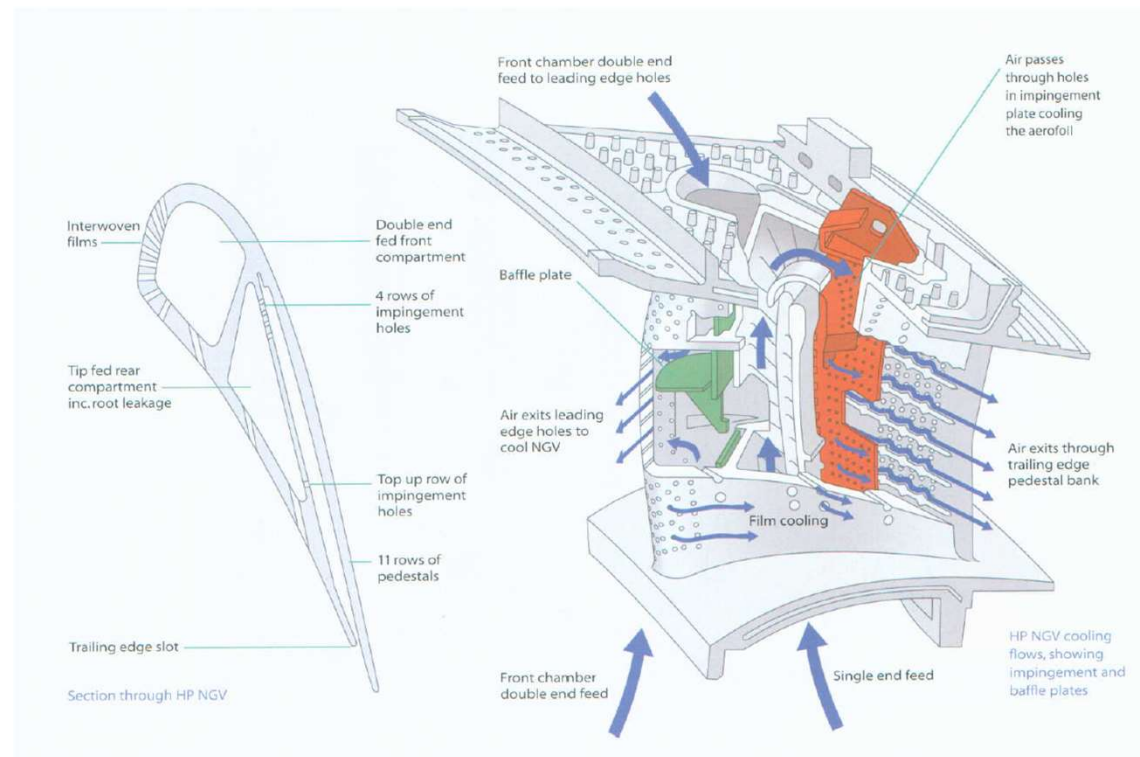
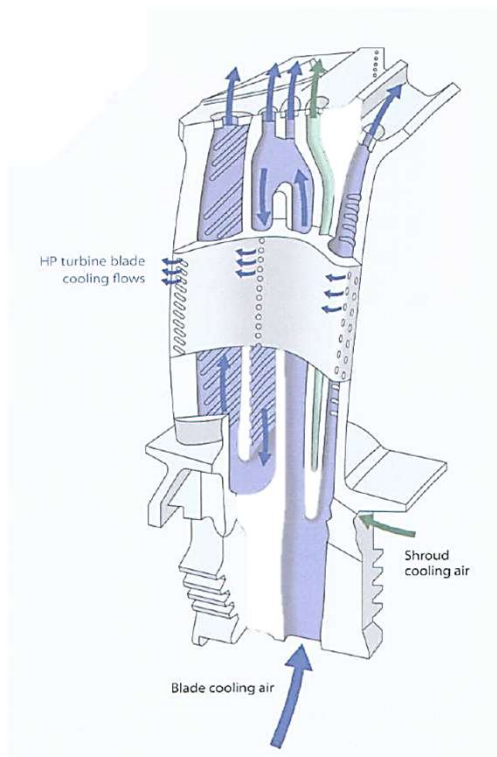
- The coefficient r in the above equations is known as the recovery factor. The recovery factor is a function of gas property, known as Prandtl number, and the state of the boundary layer, following

$$r = \sqrt{Pr} \text{ for (flat plate) laminar boundary layers}$$

$$r = \sqrt[3]{Pr} \text{ for a flat plate turbulent boundary layers}$$

- where Pr is the gas Prandtl number

Turbine Cooling Design



Example 10.5

- The total gas temperature at the exit of a turbine nozzle is $T_{tg} = 1700$ K and the local gas Mach number is $M = 1.0$. The gas ratio of specific heats is $\gamma = 1.33$, $c_p = 1157$ J/kg · K, and Prandtl number is $Pr = 0.71$. First calculate:
 - a) the gas static temperature T_g
 - b) the adiabatic wall temperature T_{aw} on the nozzle for a turbulent boundary layer
 - c) the adiabatic wall temperature on the nozzle for a laminar boundary layer
- Assuming that the gas speed relative to the rotor is $W_2 = 455$ m/s, calculate
 - d) the total temperature of the gas on the rotor, i.e., as measured in the rotor frame of reference, $T_{t,r}$.

Turbine Cooling Mass Fraction

- Setting the heat transfer rate on the gas side and the coolant side equal to each other and expressing the ratio of the coolant to gas flow rate as ε , the expression for the coolant mass fraction is

$$\varepsilon \equiv \frac{\dot{m}_c}{\dot{m}_g} \equiv \frac{C_{p_g} St_g}{C_{p_c} St_c} \left(\frac{T_{aw} - T_{wg}}{T_{wc} - T_c} \right)$$

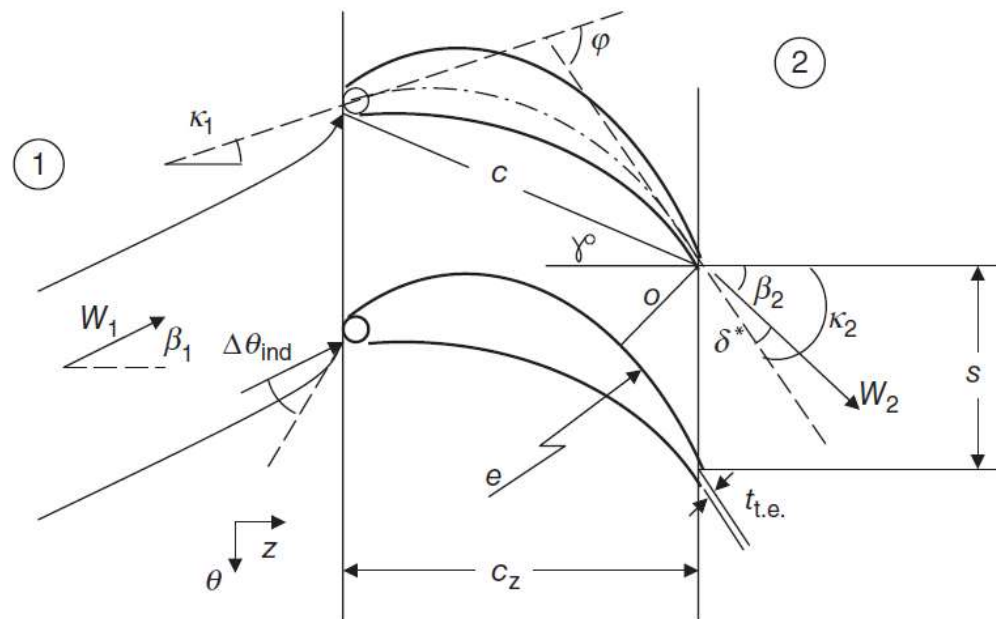
- To see a correlation between the convective film coefficients (h_g and/or h_d) for the gas and the coolant flow respectively, the flow Reynolds number and Prandtl number we may write the heat transfer rate relation from the hot gas and the coolant using Stanton number.

The Effect of Cooling on Turbine Efficiency

- The impact of cooling on turbine efficiency may be attributed to the following effects:
 - The coolant mass flow rate does not participate in turbine power production; therefore, per 1% cooling fraction there is ~1% loss in power production (due to loss of working fluid)
 - The coolant injection in the hot gas stream causes mixing losses of two streams as well as an increase in profile drag loss on the blades (for disruption of flow on the blades, as in film holes)
 - The coolant suffers a total pressure drop inside the cooling passages due to friction, turbulators, pins, and other roughness elements inside the cooled turbine blades. The reduced total pressure on the part of the coolant then causes the stage total pressure ratio π_t to drop
 - The heat transfer between the hot gas and the coolant causes an entropy rise for the mixed-out gas.
- The turbine efficiency may be defined as the ratio of actual turbine work per total airflow (that includes the coolant fraction) and the ideal turbine work, achieved isentropically, across the actual turbine expansion, $(p_5/p_4)_{\text{actual}}$.

Turbine Blade Profile Design

- The basic parameters are the same as the compressor cascade, some of the distinguishing features in a turbine cascade, are an inlet induced flow angle $\Delta\theta_{ind}$, or throat opening o , or the suction surface curvature e , downstream of the throat, and finally the trailing-edge thickness $t_{t.e.}$.



Turbine Blade Profile Design

- Angles
 - In order to construct a suitable turbine blade profile, we need to estimate the inlet and exit blade angles and their relation to the actual incidence and deviation angles. The incidence angle in a turbine cascade accounts for the flow curvature near the leading edge, called the *induced turning*, $\Delta\theta_{ind}$ and is called *actual incidence* i_{ac} . The correlation between the induced angle, inlet flow angle, and blade solidity is

$$\Delta\theta_{ind} = 14 \left(1 - \frac{\beta_1}{70^\circ} \right) + 9(1.8 - \sigma)$$

- The actual incidence and flow angles are corrected by the induced angle according to

$$i_{ac} = i + \Delta\theta_{ind}$$

$$\beta_{ac} = \beta_1 + \Delta\theta_{ind}$$

- The flow and turbine blade angles at the leading edge follow the same relation as the compressor, that is,

$$i = \beta_1 - k_1$$

Turbine Blade Profile Design

- Angles
 - The deviation angle is also important to the turbine profile design, as it adds to the blade camber, and if it is underpredicted, the exit swirl will be less than the design value and thus blade torque and in case of rotor, power, will be less than expected. Carter's rule for deviation angle in a turbine, although not the most accurate, is adequate for the preliminary design purposes,

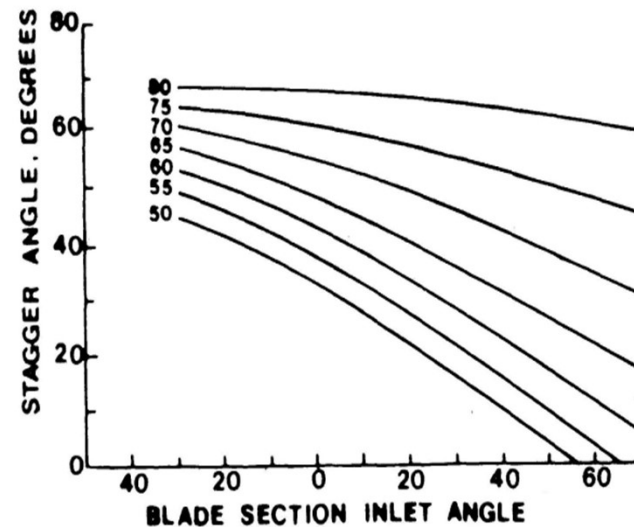
$$\delta^* = \frac{m\Delta\phi}{\sigma}$$

Turbine Blade Profile Design

- Angles
 - Stagger, or blade setting, angle is critical to the *smoothness* of turbine flow passage (area distribution) design. The simple approximation equates the stagger to the mean flow angle in a blade row, that is,

$$\gamma^\circ \approx \beta_m \approx \tan^{-1}(W_{\theta m}/C_z) \text{ (rotor)}$$

$$\gamma^\circ \approx \alpha_m \approx \tan^{-1}(W_{\theta m}/C_z) \text{ (nozzle)}$$



A more accurate determination of the stagger angle, based on the blade leading and trailing-edge angles, κ_1 and κ_2

Turbine Blade Profile Design

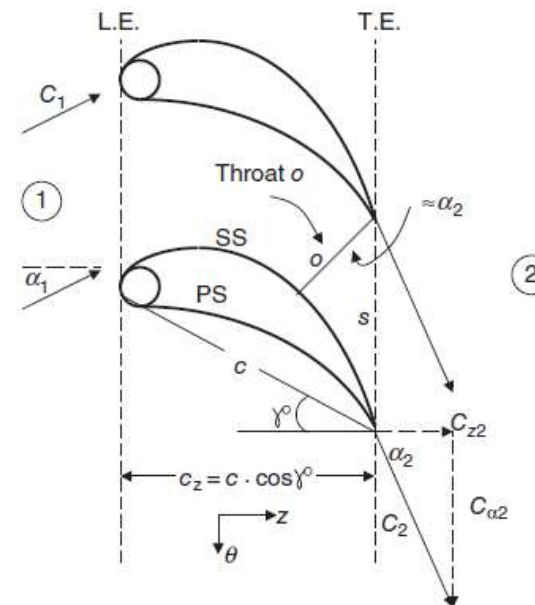
- Throat Sizing

- The throat sizing in a turbine nozzle (or rotor) is very important both for choked and unchoked nozzles. The geometry that is shown in the following definition sketch is used to relate the throat width or opening o to the blade spacing s .
- The throat opening o is related to the spacing and cosine of the exit flow angle α_2 in nozzle and β_2 in rotor, following

$$o \approx s \cdot \cos \alpha_2$$

- This approximation is acceptable for the subsonic exit flow, however, for the supersonic exit Mach numbers (but below 1.3), we correct the throat opening by the inverse of A/A^* corresponding to the supersonic exit Mach number, that is,

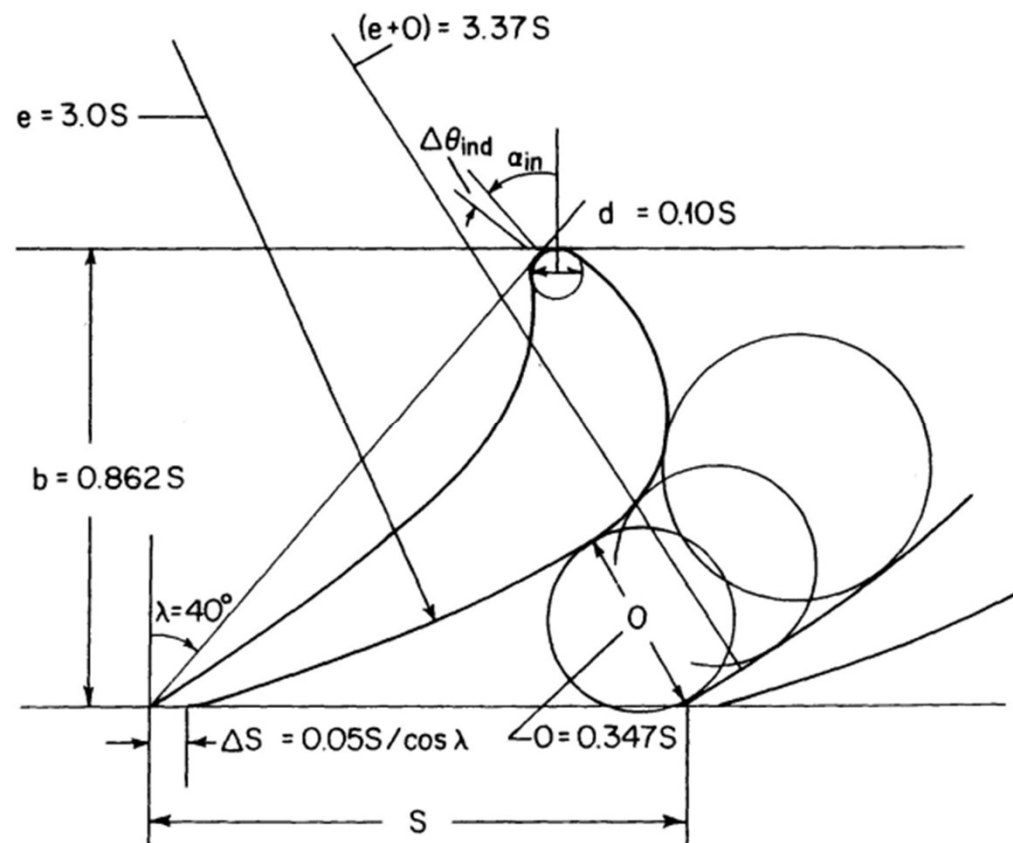
$$\frac{o}{s} \approx \frac{\cos \alpha_2}{(A/A^*) M_2}$$



Turbine Blade Profile Design

Design parameters

1. stagger angle $\gamma^\circ = 40^\circ$
2. axial chord $c_z/c = \cos \gamma^\circ = 0.766$,
3. solidity, $c/s = 1.125$,
4. $\Delta\theta_{ind} = 11.62^\circ$,
5. flow inlet angle $\alpha_{in} = +42.27^\circ$,
6. flow outlet angle $\alpha_{out} = -69.86^\circ$,



Gas Turbine Design Guidelines

We used some commonly accepted design practices, for example, constant axial velocity, to design suitable velocity triangles at the pitchline. In the process, we encountered some *hard* and *soft* design constraints. Some examples of hard and soft design constraints are summarized below:

Hard design constraints

$M_2 > 1$ (choked first nozzle)

$M_{3r} < 1$ (unchoked rotor exit flow)

$M_{5r} < 1$ (unchoked rotor exit flow)

T_{wg} (maximum wall temperature)

$^{\circ}R > 0$ (positive degree of reaction)

Soft Design Constraints

$\sigma \neq \sigma_{opt}$ (solidity other than the optimum)

$m \neq \text{constant}$ (pitchline radius is variable)

$\alpha_{exit} \neq 0$ (turbine exit swirl not zero)

$t_{wall} = 2 \text{ mm}$ (wall thickness other than nominal)

$0 \leq ^{\circ}R \leq 1$ (wide-ranging choice of $^{\circ}R$)

$\psi > 1$ (higher loading than 1 is acceptable)

$Cz \neq \text{constant}$ (acceptable)

Gas Turbine Design Guidelines

